

- (a) Suppose you are given a triangulation with v vertices, h of which lie on the outer face. Use Euler's formula to derive the number of triangles t and the number of edges e in the triangulation as functions of v and h .

Solution: There's always one more face than there are triangles because of the external face.

$$f = t + 1$$

Euler's formula:

$$v - e + f = 2$$

$$v - e + t + 1 = 2$$

$$v - e + t = 1$$

$$2v - 2e + 2t = 2$$

Total number of edges incident to all triangles:

$$3t = 2e - h$$

$$2e = 3t + h$$

Replace $2e$ with $3t + h$ in the equation $2v - 2e + 2t = 2$:

$$2v - 3t - h + 2t = 2$$

$$2v - t - h = 2$$

$$t = 2v - h - 2$$

Plug t back into $v - e + t = 1$:

$$v - e + 2v - h - 2 = 1$$

$$3v - e - h = 3$$

$$e = 3v - h - 3$$

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- (b) Suppose you are given a quadrangulation with v vertices, h of which lie on the outer face. Use Euler's formula to derive the number of quadrangles q and the number of edges e in the quadrangulation as functions of v and h .

Solution: Total number of edges incident to all quadrangles:

$$4q = 2e - h$$

$$2e = 4q + h$$

Euler's formula (skipping steps shown in (a)):

$$v - e + q = 1$$

$$2v - 2e + 2q = 2$$

Replace $2e$ with $4q + h$ in the equation $2v - 2e + 2q = 2$:

$$2v - 4q - h + 2q = 2$$

$$2v - 2q - h = 2$$

$$2q = 2v - h - 2$$

$$q = v - \frac{h}{2} - 1$$

Plug q back into $v - e + q = 1$:

$$v - e + v - \frac{h}{2} - 1 = 1$$

$$2v - e - \frac{h}{2} = 2$$

$$e = 2v - \frac{h}{2} - 2$$

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- (c) Explain why your answer to part (b) implies that no quadrangulation exists where the number of vertices on the outer face is odd.

Solution: If h was odd, then $\frac{h}{2}$ would not be an integer, therefore both the number of quadrangles and the number of edges would not be integers. We couldn't have a partial quadrangle or edge though so h must not be odd. ■

- (a) Compute the line equations of the two "crossing tangents", that is, the lines $l_1 : y = a_1x - b_1$ and $l_2 : y = a_2x - b_2$ that are both supporting lines for the convex hulls of each of P and Q such that P lies below l_1 and above l_2 and the reverse holds for Q . (Note that you are not given the hulls, just the point sets.) Your algorithm should run in $O(n + m)$ time.

Solution: ■

- (b) You have a cannon in $\mathbb{R}^2$. It has three controls labeled "a", "b", and "c". A projectile shot from this cannon travels along the parabolic arc $y = a + bx - cx^2$. You are asked to determine whether it is possible to adjust the controls so that the projectile travels above a set of n building tops, represented by a point set $P = \{p_1, \dots, p_n\}$ and beneath a set of m floating balloons, represented by a point set $Q = \{q_1, \dots, q_m\}$. Your algorithm should run in time $O(n + m)$. (Do not worry about the cannon's location. Just determine if there is some parabola along which the projectile can travel.)

Solution: ■

- (c) You are given a set of n halfplanes $H = \{h_1, \dots, h_n\}$. Each halfplane h_i is given as a pair (a_i, b_i) and consists of all the points (x, y) such that $y \leq a_ix - b_i$. Compute the axis-parallel square of the largest side length lying in $h_1 \cap \dots \cap h_n$ whose lower edge lies on the x-axis. If no such square exists, your algorithm should indicate this.

Solution: ■

- (a) Suppose we are given n points and create a subdivision as described above. Show that the resulting subdivision has size $O(n)$ (counting vertices, edges, and faces).

Solution: Each time we add a point to the subdivision, there are three possible cases:

- (a) The point is inside an existing face. In this case, we shoot bullets left, right, and up. Each of these creates a new vertex when it hits an edge. Add in the point itself and that's 4 new vertices. Each bullet creates a new edge so 3 new edges. There's now a face below the horizontal line and to the left and right of the vertical line. This adds 2 faces.

So in total we add 4 vertices, 3 edges, and 2 faces.

- (b) The point is on a horizontal edge. In this case, we split the edge into two edges and shoot a bullet up. The face is split into two faces. One new vertex is created at the end of the bullet path and a vertex is added for the new point.

So in total we add 2 vertices, 2 edges, and 1 face.

- (c) The point is on a vertical edge. In this case, we split the edge into two edges and shoot a bullet left and right. The left and right faces are split into two faces each. One new vertex is created at the end of each bullet path and a vertex is added for the new point.

So we add 3 vertices, 3 edges, and 2 faces.

In the worst case, we add 4 vertices, 3 edges, and 2 faces for each point. So the size of the subdivision is $(4 + 3 + 2) \cdot n = 9n = O(n)$. ■

- (b) Prove that for any $1 \leq i \leq n$, the expected number of new boxes created during the i th addition is $O(1)$.

Solution: Pulling heavily from Mount's notes because this is basically the same as the proof that the expected number of new trapezoids created during the i th addition to a trapezoidal map is $O(1)$.

Lemma: Consider the randomized incremental construction of a subdivision given points $P = \{p_1, \dots, p_n\}$, and let k_i denote the number of new boxes created when the i th point is added. Then $E[k_i] = O(1)$, where the expectation is taken over all possible permutations of the points as the insertion orders.

Proof: Let S_i denote the subdivision resulting from the insertion of the i th point. Because we are averaging over all permutations, among the i points in S_i , each point has an equal probability $\frac{1}{i}$ of being the last one added. For each of the points p , we want to count the number of boxes that would have been created if p had been the last point added (The worst case scenario for p).

We say that a box b depends on a point p if p would have caused b to be created had p been the last point inserted. We want to count the number of boxes that depend on each point and then compute the average over all points. If we let $\delta(b, p) = 1$ if b depends on p and 0 otherwise, then the expected value is:

$$E[k_i] = \sum_{p \in P_i} (\text{probability that } p \text{ is last}) \cdot (\text{number of boxes that depend on } p)$$

Where P_i is the set of points we've considered so far. We can rewrite this as:

$$\begin{aligned} E[k_i] &= \sum_{p \in P_i} \frac{1}{i} \left(\sum_{b \in S_i} \delta(b, p) \right) \\ &= \frac{1}{i} \sum_{p \in P_i} \sum_{b \in S_i} \delta(b, p) \\ &= \frac{1}{i} \sum_{b \in S_i} \sum_{p \in P_i} \delta(b, p) \end{aligned}$$

Each box is bounded by lines from at most 4 points. It may be less than 4 if the box is bounded by the bounding rectangle instead. The top and bottom of each box are formed by bullet paths going left or right from two points (or by the boundary rectangle of course). The left and right are formed by bullet paths going up from two points.

Therefore, $\sum_{p \in P_i} \delta(b, p) \leq 4$.

Since S_i consists of at most $2i + 1$ boxes (we start with 1 box and, as stated in part (a), add at most 2 boxes for each point), we have:

$$E[k_i] \leq \frac{1}{i} \sum_{b \in S_i} 4 = \frac{4}{i} |S_i| \leq \frac{4}{i} (2i + 1) = O(1)$$

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